

This paper aims to measure the potential *seniority-biased* impact of GenAI on the labor market. Specifically, we ask whether GenAI adoption by firms disproportionately affects junior roles relative to more senior positions. This perspective extends the classic literature on skill-biased technological change (e.g., [Katz and Murphy, 1992](#); [Autor et al., 2003](#); [Acemoglu and Autor, 2011](#)), which emphasizes shifts in labor demand across education or occupation groups, to a related but distinct dimension: *seniority*.³

We begin with a simple conceptual framework, building on [Acemoglu and Autor \(2011\)](#) and [Acemoglu and Restrepo \(2018\)](#), to analyze why GenAI adoption may disproportionately reduce demand for junior workers and to clarify the mechanisms underlying this pattern. The framework is motivated by two broad patterns in the emerging evidence: micro-level experimental studies consistently find that GenAI disproportionately increases the productivity of less-experienced and less-skilled workers in a given task ([Brynjolfsson et al., 2025b](#); [Noy and Zhang, 2023](#); [Dell’Acqua et al., 2023](#); [Cui et al., 2025](#); [Cruces et al., 2026](#)), whereas labor market evidence, including our results, links GenAI adoption and exposure to declines in junior employment ([Brynjolfsson et al., 2025a](#); [Azar et al., 2025](#); [Klein Teeselink, 2025](#)). The framework reconciles these patterns.

In the framework, junior and senior workers sort into a continuum of tasks according to comparative advantage. GenAI both automates low-complexity tasks previously performed by juniors and augments productivity on remaining tasks, with larger gains for juniors. The change in junior labor demand decomposes into three forces: a *displacement effect*, capturing the loss of automated tasks; a *productivity effect*, whereby higher productivity reduces labor required per task; and a *reallocation effect*, whereby juniors expand into some previously senior tasks as their relative productivity rises (in the spirit of [Althoff and Reichardt, 2026](#); [Hosseini and Lichtinger, 2026](#)). When tasks are complementary in production, both displacement and productivity effects reduce junior labor demand, while reallocation is the only offsetting force and depends on the extent of differential augmentation. Whether junior employment declines more than senior employment therefore depends on which force dominates.

We then move to our main empirical analysis, drawing on a dataset that combines LinkedIn résumés and job-posting data from Revelio Labs ([Revelio Labs, 2025](#)). The

³See [Ide and Talamas \(2025\)](#) for a closely related theoretical framework, analyzing how GenAI may differentially affect workers within firms depending on their knowledge and role.

to senior employment. Before turning to this aggregate evidence, we first examine the framework’s underlying mechanisms. Using an LLM, we map a representative sample of our job postings to standardized O*NET tasks and show that AI-exposed tasks become disproportionately less likely to appear in junior task bundles at adopting firms after 2023, while no comparable pattern emerges for seniors. This provides evidence consistent with the displacement channel. By contrast, tasks that were ex ante concentrated among seniors do not systematically migrate into junior task bundles, offering little support for the reallocation channel. We complement these tests with external task-level data from the Anthropic Economic Index (Appel et al., 2026), documenting that real-world GenAI use is more likely to take the form of automation for tasks characteristic of junior occupations, whereas usage in senior occupations is more often augmentative, consistent with the framework’s assumption that junior tasks are more susceptible to displacement.

We then turn to the firm-level employment response. We first compare junior and senior employment at adopting and non-adopting firms. From 2018 through 2022, junior employment evolves similarly in the two groups. Beginning in late 2022, however, junior employment at adopters diverges sharply downward relative to non-adopters, while senior employment shows no comparable break. To directly assess the seniority-biased effect, we estimate a triple-difference specification that compares junior versus senior employment within adopting firms relative to non-adopters. This design includes firm-by-time fixed effects, which absorb shocks or trajectories specific to a given firm in a given period, and industry-by-time-by-seniority fixed effects, which account for sector-level dynamics that differentially affect juniors and seniors. The coefficients are flat through 2022Q4 and then decline sharply from 2023Q1 onward, reaching roughly a 9 percent decline in junior employment relative to senior employment after six quarters.

We next examine which types of jobs drive our results, focusing on heterogeneity by occupational exposure to GenAI using the widely used exposure measure of Eloundou et al. (2024). Specifically, we estimate a triple-difference specification that compares employment changes in high- versus low-exposure occupations within adopting and non-adopting firms, separately for junior and senior workers. Through the end of 2022, the estimated coefficients for juniors and seniors follow nearly identical trajectories. Beginning in 2023Q1, however, the two series diverge sharply. Junior employment in high-exposure occupations declines by roughly 7 percent relative to low-exposure occupations over the

Table 4: Hiring, Separations, and Promotions

	Hiring	Separation	Promotion (rate)
Treat $\times$ Post	−4.006*** (0.221)	−1.089*** (0.163)	0.033** (0.014)
Pre-period mean	5.059	4.154	0.822
Observations	7,941,179	7,941,179	7,690,581
Clusters (firms)		280,866	

Notes: This table reports the estimated β from Equation 15. Estimation begins in 2018 to exclude early-sample trends. Standard errors clustered by firm in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6 Conclusion

This paper provides early, large-scale evidence that the diffusion of GenAI since 2023 is associated with *seniority-biased* employment effects within firms. Using résumé and job-posting data linked to roughly 281,000 U.S. firms, together with a firm-level adoption measure based on “GenAI integrator” vacancies, we document a decline in junior employment relative to senior employment at adopting firms compared with non-adopting firms. This decline is concentrated in high-exposure occupations: within adopting firms, junior employment in high-exposure occupations contracts relative to low-exposure occupations after 2022, while the corresponding senior series continues to rise. Task-level evidence from job postings points to the same mechanism: GenAI-exposed tasks become less likely to appear in junior task bundles at adopting firms. Finally, decomposing employment flows shows that the decline is driven primarily by slower hiring rather than increased separations.

These findings should be interpreted with caution. Adopting firms differ systematically in size, workforce composition, and industry. Although our research designs absorb many observable and unobservable differences across firms, sectors, and time, unobserved confounding factors may remain. We therefore examine four prominent alternative explanations: monetary tightening, post-pandemic hiring dynamics, the expansion of remote work, and labor-supply shocks. These tests support our interpretation, but they cannot rule out all alternative explanations. In addition, our adoption measure captures deliberate organizational integration of GenAI but may miss informal or “silent” adop-

tion within firms. Finally, because GenAI is still in the early stages of diffusion, longer-run adjustments in training, task allocation, wages, and internal career ladders may either attenuate or amplify the effects documented here.

Even with these caveats, the results suggest that GenAI adoption may be shifting labor demand away from entry-level tasks and narrowing the lower rungs of internal career ladders. Because early-career jobs are central to skill formation and lifetime earnings growth, such changes may have consequences beyond short-run employment. Whether these patterns persist, and whether firms respond by redesigning training, tasks, or career progression, remains an important question for future research.

References

- Acemoglu, D. and Autor, D. (2011). Skills, Tasks and Technologies: Implications for Employment and Earnings. In *Handbook of Labor Economics*, volume 4, pages 1043–1171. Elsevier.
- Acemoglu, D., Autor, D., Hazell, J., and Restrepo, P. (2022). Artificial Intelligence and Jobs: Evidence from Online Vacancies. *Journal of Labor Economics*, 40(S1):S293–S340.
- Acemoglu, D. and Restrepo, P. (2018). The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment. *American Economic Review*, 108(6):1488–1542.
- Acemoglu, D. and Restrepo, P. (2019). Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives*, 33(2):3–30.
- Acemoglu, D. and Restrepo, P. (2020). Robots and Jobs: Evidence from U.S. Labor Markets. *Journal of Political Economy*, 128(6):2188–2244.
- Acemoglu, D. and Restrepo, P. (2022). Tasks, Automation, and the Rise in US Wage Inequality. *Econometrica*, 90(5):1973–2016.
- Althoff, L. and Reichardt, H. (2026). Task-Specific Technical Change and Comparative Advantage.
- Appel, R., Massenkoff, M., McCrory, P., McCain, M., Heller, R., Neylon, T., and Tamkin, A. (2026). The Anthropic Economic Index report: Economic Primitives.
- Autor, D. (2024). Applying AI to Rebuild Middle Class Jobs. Technical report, National Bureau of Economic Research.
- Autor, D., Levy, F., and Murnane, R. J. (2003). The Skill Content of Recent Technological Change: An Empirical Exploration. *The Quarterly Journal of Economics*, 118(4):1279–1333.
- Autor, D. H. and Dorn, D. (2013). The Growth of Low-Skill Service Jobs and the Polarization of the U.S. Labor Market. *American Economic Review*, 103(5):1553–1597.
- Azar, J., Gine, M., and Sanz-Espín, J. (2025). AI Is Already Eroding Wages: Quasi-Experimental Evidence From Occupational Exposure. Technical report.
- Babina, T., Fedyk, A., He, A., and Hodson, J. (2024). Artificial Intelligence, Firm Growth, and Product Innovation. *Journal of Financial Economics*, 151:103745.

- Financial Times (2025). Is AI Killing Graduate Jobs? By Clara Murray, Delphine Strauss, John Burn-Murdoch, and Sarah Lim. Published July 24, 2025.
- Forbes (2025). As AI Reduces New Grad Hiring, Apprenticeships Will Become Essential.
- Friebel, G., Huang, Y., Li, J., Shukla, S., and Zhang, A. (2026). Pyramids, Diamonds, and Oscillations: AI and the Structure of Internal Labor Markets.
- Garicano, L. (2000). Hierarchies and the Organization of Knowledge in Production. *Journal of Political Economy*, 108(5):874–904.
- Garicano, L. and Rayo, L. (2025). Training in the Age of AI: A Theory of Career Viability. Technical report, Centre for Economic Policy Research.
- Garicano, L. and Rossi-Hansberg, E. (2006). Organization and Inequality in a Knowledge Economy. *Quarterly Journal of Economics*, 121(4):1383–1435.
- Gertler, M. and Gilchrist, S. (1994). Monetary Policy, Business Cycles, and the Behavior of Small Manufacturing Firms. *Quarterly Journal of Economics*, 109(2):309–340.
- Guvenen, F., Kaplan, G., Song, J., and Weidner, J. (2022). Lifetime Earnings in the United States over Six Decades. *American Economic Journal: Applied Economics*, 14(4):446–479.
- Hampole, M., Papanikolaou, D., Schmidt, L. D., and Seegmiller, B. (2025). Artificial Intelligence and the Labor Market. Technical report, National Bureau of Economic Research.
- Hershbein, B. and Kahn, L. B. (2018). Do Recessions Accelerate Routine-Biased Technological Change? Evidence From Vacancy Postings. *American Economic Review*, 108(7):1737–1772.
- Hosseini, S. M. and Lichtinger, G. (2026). Generative AI and Occupational Entry Barriers: The Labor-Supply Channel of Technological Change.
- Houston, D. (2023). A message from our CEO. *Dropbox Blog*. Accessed 2025-10-29.
- Humlum, A. and Vestergaard, E. (2025). Large Language Models, Small Labor Market Effects. Technical report, National Bureau of Economic Research.
- Ide, E. (2025). Automation, AI, and the intergenerational transmission of knowledge. *arXiv preprint arXiv:2507.16078*.
- Ide, E. and Talamas, E. (2025). Artificial Intelligence in the Knowledge Economy. *Journal of Political Economy*. Forthcoming.
- Ilut, C., Kehrig, M., and Schneider, M. (2018). Slow to hire, quick to fire: Employment dynamics with asymmetric responses to news. *Journal of Political Economy*, 126(5):2011–2071.
- Intelligent (2024). 8 in 10 Companies Plan to Layoff Recent College Grads This Year Due to AI. Survey of 804 U.S. hiring managers conducted via Pollfish, April 2024.
- Jarocinski, M. and Karadi, P. (2020). Deconstructing Monetary Policy Surprises—The Role of Information Shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43.
- Jordà, O. (2005). Estimation and Inference of Impulse Responses by Local Projections. *American Economic Review*, 95(1):161–182.
- Katz, L. F. and Murphy, K. M. (1992). Changes in Relative Wages, 1963–1987: Supply and Demand Factors. *The Quarterly Journal of Economics*, 107(1):35–78.
- Klein Teeselink, B. (2025). Generative AI and labor market outcomes: Evidence from the United Kingdom. Technical report.
- Lambert, P. J. and Schindler, Y. (2026). The Broken Ladder: AI, Remote Work, and Early-Career Hiring. SSRN.